

UUM Student Satisfaction Analysis Monitoring Tool

Requirements Specification

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1. INTRODUCTION

This section introduces the requirement specification document that specifies all system requirements that will govern the development and implementation of the UUM Student Satisfaction Analysis Monitoring Tool. It provides the purpose and scope of the system as well as the definition of terms specific to the system.

1.1 Purpose

This "UUM Student Satisfaction Analysis Monitoring Tool" offers some contributions. The sentiment analysis process enables stakeholders to quickly collect, extract, analyze, and present actual data results, alleviating the resources and time required to gather information. Secondly, it helps stakeholders understand the analysis results and implement the necessary solutions to help students solve real-life problems. For example, suppose they want to know how students live in the dormitory. In this case, they can enter and search for posts about the student environment and then extract and analyse those posts to decide whether to improve or maintain the status. Thirdly, the stakeholders have the opportunity to understand and learn how to use sentiment analysis to improve their knowledge of managing the university. Finally, it also helps to change the perception of people (not only teachers, students, and staff) about the university in terms of management.

1.2 Scope

The scope of this project are:

1. "UUM Student Satisfaction Analysis Monitoring Tool" will be developed for the management authority of each department in UUM to analyze and visualize the collected data from social media.
2. The data sources will be from a social media group, which is N.E.W.S.E.E.D., on Facebook.
3. The post by students is chosen to focus on this project. Posts like announcements or advertisements will be eliminated.
4. The posts written in languages other than English, such as Malay, Mandarin, and Tamil are removed.
5. The positive and negative sentiments of the key or particular words about UUM students' opinions will visualize on the monitoring tool.

1.3 Definitions, Acronyms and Abbreviations

UUMSSMAT- UUM Student Satisfaction Analysis Monitoring Tool

2. GENERAL DESCRIPTION

This section provides a high-level description of the UUM Student Satisfaction Analysis Monitoring Tool, identifies the users involved and their roles, and specifies the general constraints, assumptions and dependencies.

2.1 System Functions

UUM Student Satisfaction Analysis Monitoring Tool is a monitoring tool that allows the management authority of each department in UUM to analyse the sentiment of Facebook posts or comments, whether it is positive or negative or neutral. To confirm the accuracy of the sentiment in the sentences, the accuracy of determining sentiment groups such as positive, negative, and neutral will be displayed in the form of percentages. Therefore, the users can know the sentiment result of the post or comment. Moreover, the user can also view the details of the sentences.

2.2 User

The user who uses this system is the management authority of each department in UUM.

2.3 General Constraints

The system requires devices and an internet connection to access the system efficiently.

2.4 Assumptions and Dependencies

Assumption:

The user needs the basic skill of using the device.

3. REQUIREMENTS

This section lists all functional and non-functional requirements for the UUM Student Satisfaction Analysis Monitoring Tool, the use case(s) model, and a flowchart.

3.1 Functional Requirements

Table 1. The priority of the requirements is indicated by:

M- Mandatory

D – Desirable.

O- Optional

Table 1: The Functional Requirements Definition

Num	Requirement ID	Requirements Description	Priorit y
1	UUMSSMAT_01	INPUT POST OR COMMENT	
	UUMSSMAT_01_01	The user must input the Facebook post or comment into the monitoring tool.	M
	UUMSSMAT_01_02	The system will save the Facebook post or comment.	M
2.	UUMSSMAT_02	ANALYSE TEXT SENTIMENT	
	UUMSSMAT_02_01	The system should analyse text sentiment based on the input from Facebook posts or comments whether it is positive, negative, or neutral.	M
	UUMSSMAT_02_02	The system should save the analysed Facebook post or comment.	M
3.	UUMSSMAT_03	VIEW ACCURACY	

	UUMSSMAT_03_01	The system should display the accuracy of the sentences by sentiment group (positive, negative, and neutral).	M
4.	UUMSSMAT_04	VIEW RESULT	
	UUMSSMAT_04_01	The system should display the result of the sentences by sentiment group (positive or negative or neutral).	M
5.	UUMSSMAT_05	VIEW DETAILS	
	UUMSSMAT_05_01	The system should display the details of the result.	M

3.2 Non-Functional Requirements

Table 2. The priority of the requirements is indicated by:

M- Mandatory

D – Desirable.

O- Optional

Table 2: The Non-functional requirements

Num	Requirement ID	Requirements Description	Priorit y
1	UUMSSMAT_05	SECURITY	
	UUMSSMAT_05_01	The system should protect the information of Facebook posts and comments.	M
2.	UUMSSMAT_06	RELIABILITY ISSUES	
	UUMSSMAT_06_01	The system must allow users to use it without any issues within 24 hours for single use.	M
	UUMSSMAT_06_02	The system should crash no more than once.	M
	UUMSSMAT_06_03	If the systems crash, it should behave perfectly normally when reloaded again.	M
3.	UUMSSMAT_07	USABILITY ISSUES	

	UUMSSMAT_07_01	The system must provide a text box to allow the users to input the Facebook post and comments.	M
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3.3 Use Case Model

A use case model for the system consists of user and UUM Student satisfaction Analysis Monitoring Tool use cases namely input post or comment, analyse text sentiment, view accuracy, view result , and view detail as illustrated in Figure 1.

Figure 1: The Use Case Diagram

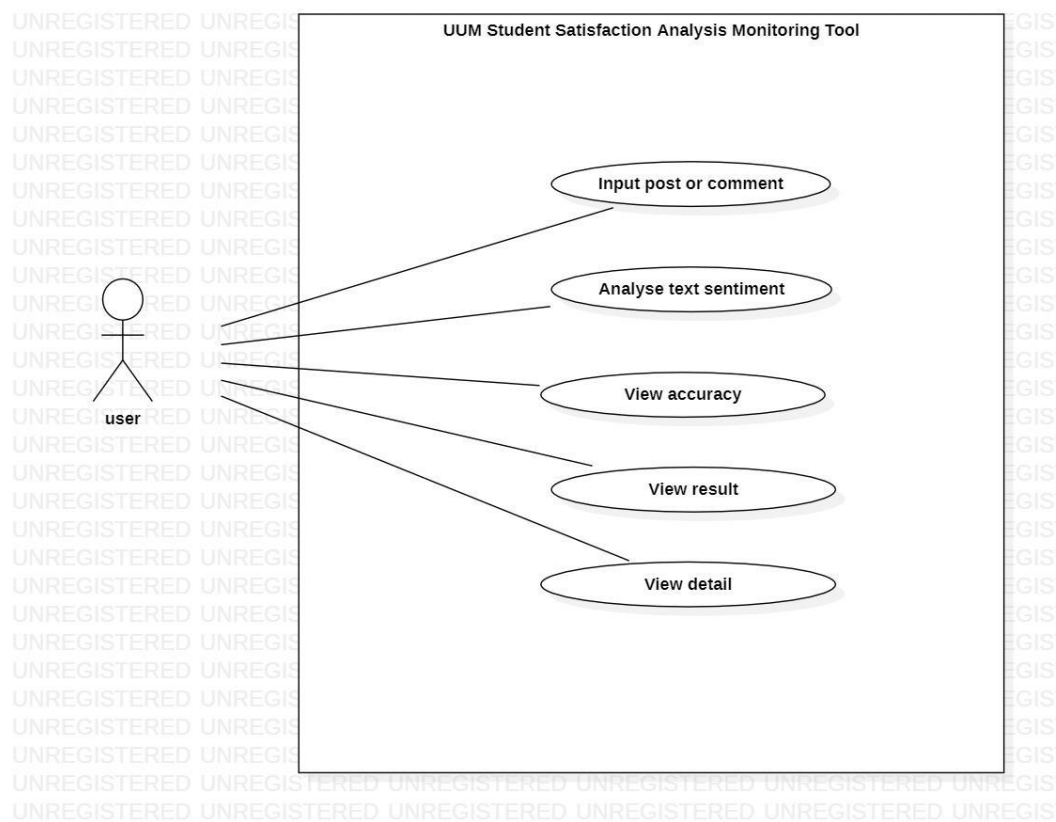
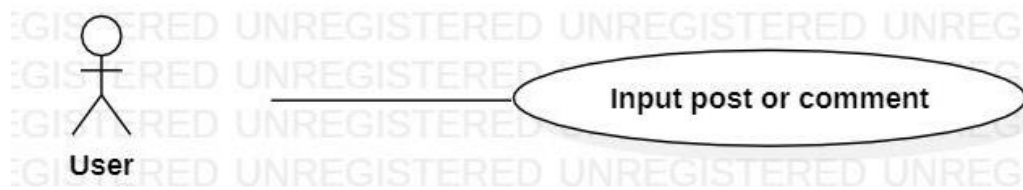
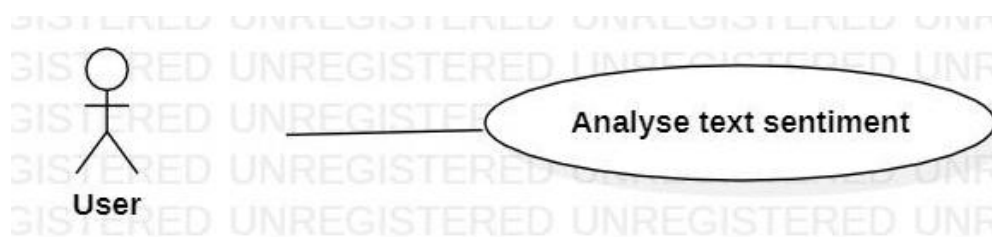


Table 3. Use case specifications

Brief Description	This use case enables the user to input the Facebook post or comment.
Pre-Conditions	Not applicable
Characteristic of Activation	Event-Driven (on user's demand)
Flow of Events	Basic Flow (UUMSSAMT_01) 1. The system must display the page 2. The user should input the Facebook post and comment in the text bar.
Post-Conditions	The system saves the information of the post and comments.
Rule(s)	Not applicable
Constraint(s)	Only one post or comment can be input at the same time.



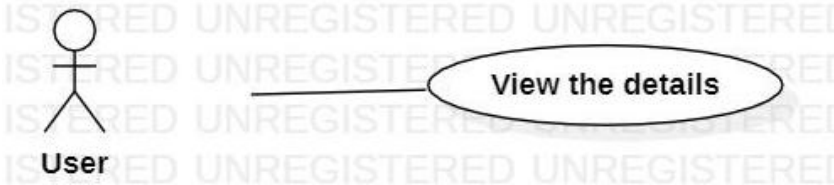
Brief Description	This use case enables the user to analyse the input Facebook post or comment, whether it is positive, negative, or neutral.
Pre-Conditions	• User must input the Facebook post or comment
Characteristic of Activation	Event-Driven (on user's demand)
Flow of Events	Basic Flow (UUMSSAMT_02) 1. The user must click the check button. 2. The system must analyse the text data based on the input from Facebook posts or comments.
Post-Conditions	The system saves the information of analysed posts and comments.
Rule(s)	Not applicable
Constraint(s)	Only one post or comment can be analysed at the same time.



Brief Description	This use case enables the user to view the accuracy of the post or comment.
Pre-Conditions	- User must input the Facebook post or comment - User must click the check button.
Characteristic of Activation	Event-Driven (on user's demand)
Flow of Events	Basic Flow (UUMSSAMT_03) The system must display the accuracy of the sentences by sentiment group (positive, negative, and neutral)
Post-Conditions	The system saves the information on the sentiment accuracy of the post and comment.
Rule(s)	Not applicable
Constraint(s)	Not applicable

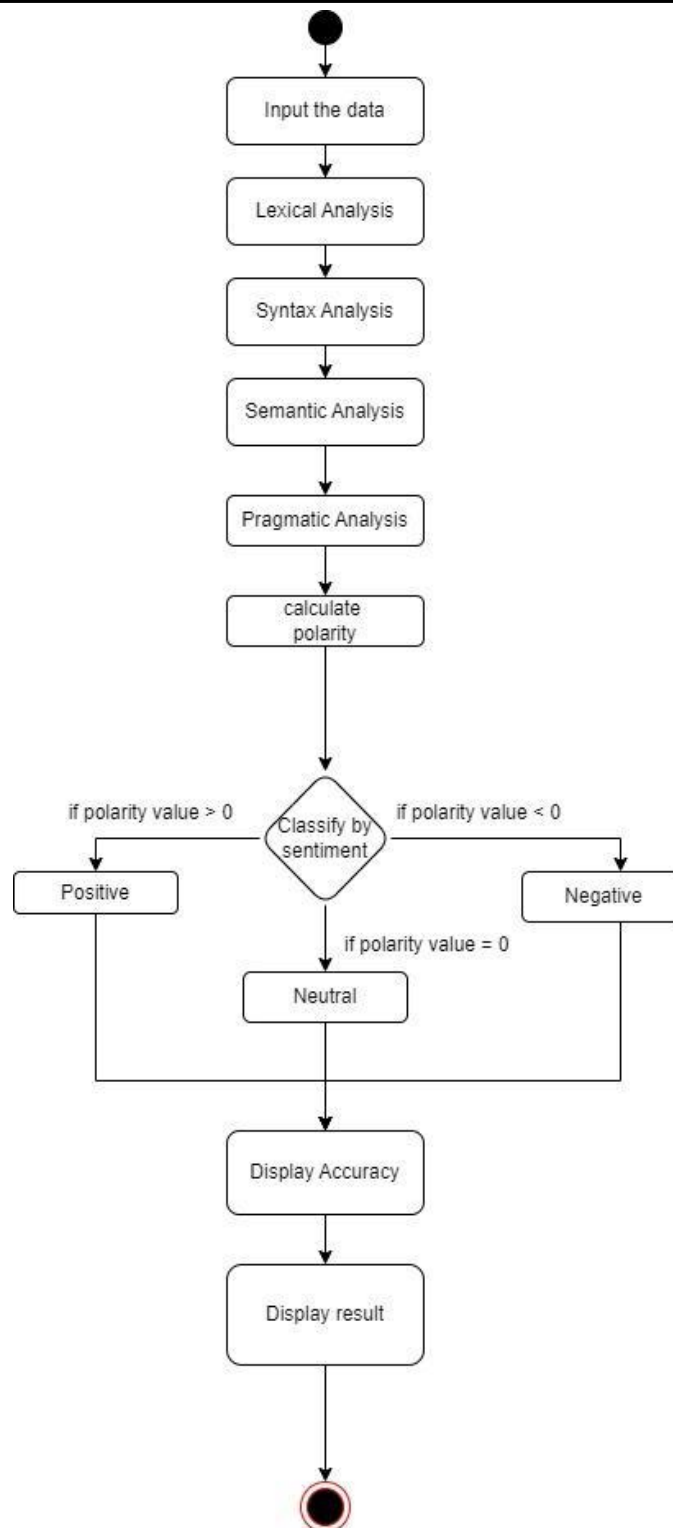


Brief Description	This use case enables the user to view the sentiment result of the post or comment.
Pre-Conditions	- User must input the Facebook post or comment - User must click the check button.
Characteristic of Activation	Event-Driven (on client's demand)
Flow of Events	Basic Flow (UUMSSAMT_04) The system must display the result of the sentences by sentiment group (positive or negative, or neutral)
Post-Conditions	The system saves the information of the sentiment result post and comment.
Rule(s)	Not applicable
Constraint(s)	Only one result is display at the same time

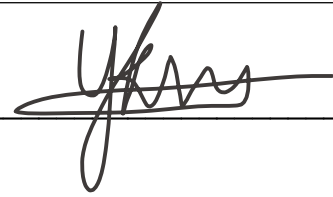


Brief Description	This use case will enable the user to view the details about the sentiment result of the post and comment.
Pre-Conditions	• User must input the Facebook post or comment • User must click the check button.
Characteristic of Activation	Event-Driven (on client's demand)
Flow of Events	Basic Flow (UUMSSAMT_05) 1. The system must display the details of the sentences in different colour by referring to the sentiment result of the post or comment. (Red for negative)
Post-Conditions	Not applicable
Rule(s)	Not applicable
Constraint(s)	Only one detail of the sentence is display at the same time.

3.4 Flow Chart



Student's Signature: _____



Date : _____

11 / 1 / 2023

TO BE FILLED BY THE SUPERVISOR

Remark:

Signature : _____



Date : _____

11 / 1 / 2023

Name : _____

Official Stamp: